

Relationship between Iron Carbide Phases (ϵ -Fe₂C, Fe₇C₃, and χ -Fe₅C₂) and Catalytic Performances of Fe/SiO₂ Fischer–Tropsch Catalysts

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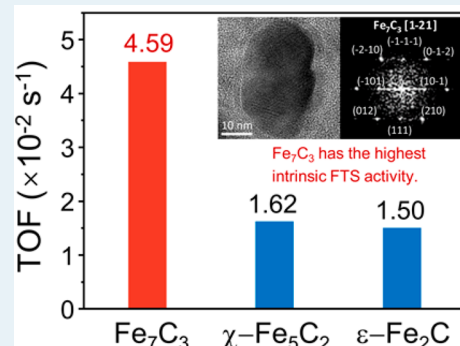
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S Supporting Information

ABSTRACT: The influence of different iron carbides on the activity and selectivity of iron-based Fischer–Tropsch catalysts has been studied. Different iron carbide phases are obtained by the pretreatment of a binary Fe/SiO₂ model catalyst (prepared by coprecipitation method) to different gas atmospheres (syngas, CO, or H₂). The phase structures, compositions, and particle sizes of the catalysts are characterized systematically by XRD, XAFS, MES, and TEM. It is found that in the syngas-treated catalyst only χ -Fe₅C₂ carbide is formed. In the CO-treated catalyst, Fe₇C₃ and χ -Fe₅C₂ with a bimodal particle size distribution are formed, while the H₂-treated catalyst exhibits the bimodal size distributed ϵ -Fe₂C and χ -Fe₅C₂ after a Fischer–Tropsch synthesis (FTS) reaction. The intrinsic FTS activity is calculated and assigned to each corresponding iron carbide based on the phase composition and the particle size. It is identified that Fe₇C₃ has the highest intrinsic activity (TOF = 4.59 × 10⁻² s⁻¹) among the three candidate carbides (ϵ -Fe₂C, Fe₇C₃, and χ -Fe₅C₂) in typical medium-temperature Fischer–Tropsch (MTFT) conditions (260–300 °C, 2–3 MPa, and H₂/CO = 2). Moreover, FTS over ϵ -Fe₂C leads to the lowest methane selectivity.

KEYWORDS: Fischer–Tropsch synthesis, iron catalyst, iron carbide, Fe₇C₃, XAFS, in situ MES



1. INTRODUCTION

Fischer–Tropsch synthesis (FTS) is a key process for converting the carbonaceous materials (coal, natural gas, and biomass) into chemicals and liquid fuels via syngas, which is ideal for the clean use of carbon-based energy.^{1,2} An iron-based catalyst is a promising system in terms of utilizing coal- and biomass-derived syngas due to its high water–gas shift (WGS) activity, flexible operating conditions, and low cost.^{3,4} However, abundant phases (Fe₂O₃, Fe₃O₄, FeO, Fe, ϵ -Fe₂C, ϵ -Fe₂C₂, Fe₇C₃, χ -Fe₅C₂, and θ -Fe₃C) of iron-based catalysts and their complicated phase transformations under reaction conditions make it difficult to identify the active phase for FTS.^{5–10}

Nowadays, it has been generally accepted that active phases are iron carbides instead of iron oxides. Single-phase χ -Fe₅C₂

and ϵ -iron carbide have been demonstrated to be active in FTS.^{11,12} θ -Fe₃C has been considered inactive and mainly appears in deactivated catalyst,^{13,14} though a recent study reported that θ -Fe₃C promoted with Mn was active for high-temperature FTO.¹⁵ Additionally, identification of Fe₇C₃ as the FTS active phase has been unsolved. Eckstrom et al.¹⁶ and Louw et al.¹⁷ observed the Fe₇C₃ phase in commercial catalyst after a long run of high-temperature fluidized FTS. In Datye's group,^{18,19} the coexistence of Fe₇C₃/α-Fe or Fe₇C₃/χ-Fe₅C₂ was observed in the working catalysts after long FTS runs.

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O'Brien et al.²⁰ prepared $\text{Fe}_7\text{C}_3/\theta\text{-Fe}_3\text{C}$ by laser pyrolysis of $\text{Fe}(\text{CO})_5$ and C_2H_4 and used it for FTS. The catalyst was more active than the ultrafine Fe_2O_3 . At similar CO conversion, the $\text{Fe}_7\text{C}_3/\theta\text{-Fe}_3\text{C}$ catalyst produced less methane and had higher C_{5+} and olefin selectivities than the precipitated catalyst. However, the carbide phases were less stable and rapidly oxidized into Fe_3O_4 during reaction. It is essential to understand the role of Fe_7C_3 in the FTS. Because the iron catalysts exhibit abundant carbide phases, it is important to identify the most active one for FTS reaction, which provides inputs for fine-tuning of iron-based FT catalysts.^{13,14,18,20–22}

Gas–solid carburization is widely used to prepare the carbide phases for FTS. Different carbide structures can be synthesized by regulating the starting materials (oxide or metal), atmosphere, temperature, particle size, etc.^{12,21–26} as these factors in essence influence the carbon chemical potential (μ_{C}).²⁷ Among them, temperature has a drastic influence on μ_{C} ²⁴ and thereby modulates the carbide phase significantly. Higher temperature produces lower μ_{C} and thus leads to carbon-deficient carbide. Furthermore, the stability sequence versus temperature for iron carbides is ϵ -carbide (below 250 °C) < $\chi\text{-Fe}_5\text{C}_2$ (250–350 °C) < $\theta\text{-Fe}_3\text{C}$ (above 350 °C). For example, $\chi\text{-Fe}_5\text{C}_2$ can be synthesized in the temperature range of 250–350 °C when treating iron oxide or metallic iron with CO or syngas.²⁴ Because the frequently adopted FTS reaction temperature (240–360 °C) is ideal for the formation of $\chi\text{-Fe}_5\text{C}_2$, it is generally believed that the active phase for FTS is $\chi\text{-Fe}_5\text{C}_2$. On the other hand, $\theta\text{-Fe}_3\text{C}$ can be prepared at a higher carburization temperature.²⁵ It is slightly hard to prepare $\epsilon\text{-Fe}_2\text{C}$ by just controlling the low temperature. Because $\epsilon\text{-Fe}_2\text{C}$ has a higher carbon content and higher space group symmetry among carbides, the formation is kinetically and entropically hindered. It is found that the starting material plays a vital role in $\epsilon\text{-Fe}_2\text{C}$ formation. For example, in order to prepare $\epsilon\text{-Fe}_2\text{C}$, iron oxide is always prereduced to $\alpha\text{-Fe}$ with H_2 and then carburized with syngas at low temperature²¹ because the permeation of carbon into $\alpha\text{-Fe}$ is easier than that into iron oxide. Xu et al.¹² successfully prepared $\epsilon\text{-Fe}_2\text{C}$ by carburizing the rapidly quenched (RQ) skeletal iron at low temperature because the structural peculiarities of the RQ Fe (low coordination number, nanoscale, and expanded lattice) are essential to overcome the seemingly insurmountable hindrance. Compared with the above carbides ($\epsilon\text{-Fe}_2\text{C}$, $\chi\text{-Fe}_5\text{C}_2$, and $\theta\text{-Fe}_3\text{C}$), it is quite a challenge to synthesize Fe_7C_3 by gas–solid carburization because of the competing formation of $\epsilon\text{-Fe}_2\text{C}$ and $\chi\text{-Fe}_5\text{C}_2$ under low or high temperature. Theoretical study suggested that Fe_7C_3 can stably exist in high μ_{C} , and a kinetic factor plays an important role in its formation.²⁴ This might account for the reason that Fe_7C_3 was mostly observed even after long FTS reactions at high temperature and high pressure. If the formation of $\epsilon\text{-Fe}_2\text{C}$ and $\chi\text{-Fe}_5\text{C}_2$ is thermodynamically or kinetically suppressed, Fe_7C_3 can be synthesized. We conceive an approach to synthesize Fe_7C_3 by treating highly dispersed ultrasmall iron oxide particles with pure CO at moderate temperature. The pure CO endows a high μ_{C} . The moderate temperature is used to mainly suppress the $\epsilon\text{-Fe}_2\text{C}$. The ultrasmall particles are favorable for carbon diffusion kinetically leading to formation of Fe_7C_3 .

A stable active catalyst composition is the prime requirement for a reliable study of FTS reaction over carbide phases. As discussed above, the stability of iron carbides is sensitive to temperature, and it is important to make sure the carbide phase will not undergo further phase changes during the FTS

conditions. For example, $\epsilon\text{-Fe}_2\text{C}$ is thermodynamically favored at low temperature (<220 °C),²⁸ but exposing it to FTS conditions (>250 °C) would make it unstable and lead to phase transformation (e.g., $\epsilon\text{-Fe}_2\text{C} \rightarrow \chi\text{-Fe}_5\text{C}_2$).²⁴ This would make the structure–performance relationship study difficult. In order to ensure the stability of a particular iron carbide phase during FTS reaction, a relatively higher temperature is used in the preparation of carbides and a lower temperature is used in FTS reaction. Besides temperature, highly dispersed active phases and a moderate carburization degree are also important factors to keep catalysts stable because the distributed active phases and limited FTS reaction can avoid local hot-spots and prevent the active particle from aggregating or sintering. It is well-known that silica is widely used as a support or binder in industrial iron-based FTS catalysts.^{19,29–32} Silica also has a stabilizing effect on iron carbides during FTS reaction.^{33,34} Specifically, only silica-supported iron catalysts were reported to observe the Fe_7C_3 phase during FTS.^{18,19} Therefore, it is suitable for a study of the relationship between iron carbides and FTS performances on moderate reduced/carburized Fe/SiO₂ catalysts.

In the present study, a stable binary Fe/SiO₂ model catalyst is used to investigate the influence of different iron carbides on the FTS activity and selectivity. The Fe/SiO₂ model catalyst was deliberately pretreated in controlled atmospheres to form different carbide phases ($\epsilon\text{-Fe}_2\text{C}$, Fe_7C_3 , and $\chi\text{-Fe}_5\text{C}_2$). The catalysts were characterized by comprehensive techniques such as XRD, XAFS, MES, and TEM. The relationship between active phases and performances is studied in detail.

2. EXPERIMENTAL SECTION

2.1. Catalyst Preparation and Controlled Synthesis of Different Iron Carbides. The catalyst precursor was prepared in a continuous coprecipitation process. A flowing aqueous solution containing $\text{Fe}(\text{NO}_3)_3$ and silica sol, with the desired iron loading of 50.9 wt % in the ultimate calcined catalyst, was mixed with flowing ammonia solution at 80 ± 1 °C and pH 8.2–8.5. The precipitate was washed and filtered with distilled water several times. The filtered cake was dried for 12 h at 120 °C and then calcined for 5 h at 500 °C in static air.

Various iron carbides were synthesized by treating the calcined catalyst with different gas atmospheres (CO, H₂, or 2H₂/CO syngas) at 300 °C and a gas space velocity of 4000 mL/(g_{cat}·h) for 20 h in a fixed-bed reactor and then subjected to FTS. These catalysts were denoted as FeSi-CO, FeSi-H₂, and FeSi-syn based on the gas used for reduction. In addition, three control experiments were also performed. Two FeSi-CO catalysts were prepared by the same procedures as before but altering only the CO exposure time to 5 and 10 h instead of 20 h. These catalysts were denoted as FeSi-CO(5h) and FeSi-CO(10h). Another catalyst was prepared from the spent FeSi-CO catalyst by reducing it with H₂ (as in the case of FeSi-H₂) and repeating the FTS reaction. This catalyst was denoted as FeSi-CO-H₂.

After FTS reaction, the reactor was transferred to a glovebox filled with N₂, and the spent catalysts were withdrawn and quickly dealt with by the following procedures. Each catalyst was divided into two parts. One part was ground to powders and loaded in a homemade sealed sample holder for XRD measurement. The other part was sealed with paraffin and then ground to powders for MES, XAFS, and TEM measurements. The reduced FeSi-CO, FeSi-H₂, and FeSi-syn catalysts before FTS were prepared in a quartz tube using the same procedures

as those in FTS evaluation. After reduction treatments with the different atmospheres, the catalysts were submerged in paraffin under gas flow to prevent oxidation.

2.2. Catalyst Characterization. Powder X-ray diffraction (XRD) spectra were obtained using a Bruker D2 X-ray diffractometer (Bruker, Germany), operating with Cu K_{α} radiation (1.54056 Å) at 30 kV and 10 mA. A continuous mode was used with a scan rate of 0.04° per second from 10 to 90°. The crystallite diameters were determined by substituting the half-width of a chosen peak into the Scherrer equation.

Mössbauer spectroscopy (MES) experiments were carried out in an MR-351 constant-acceleration Mössbauer spectrometer (FAST, Germany) driven with a triangular reference signal at 11 K. In situ MES experiments were carried out at the Reactor Institute Delft, Faculty of Applied Sciences, Delft University of Technology. The in situ reactor is similar to the one described in the book *Industrial Applications of the Mössbauer Effect*.³⁵ When the pretreatment or FTS was finished, the reactor was cooled to 300 K, and then the data were collected. Data analysis was performed using the MossWinn 4.0 software package.³⁶ The velocity was calibrated by α -Fe foil, and the isomer shift (IS) value was referenced to α -Fe. The spectra were modeled as a combination of quadrupole doublets and magnetic sextets based on a Lorentzian line shape profile. The components were identified based on their IS, quadrupole splitting (QS), and magnetic hyperfine fields (Hhf). The phase compositions were determined by the areas of the absorption peaks with the assumption that the recoil-free factor was the same for all kinds of iron nuclei in the catalyst.

Transmission electron microscopy (TEM) results were obtained on JEM-2100F (JEOL, Japan) and Talos 200A (FEI, America) electron microscopes with an acceleration voltage of 200 kV. The elemental maps were obtained by using high-angle annular dark-field scanning TEM (HAADF-STEM) of the energy-dispersive X-ray (EDX) mode. A small amount of paraffin-protected sample was dispersed in ethanol and treated by sonication for 0.5 h. Then, the suspension was transferred onto a copper grid with a holey carbon film for TEM observation.

The Fe K-edge X-ray absorption near edge spectra (XANES) and extended X-ray absorption fine structure (EXAFS) were measured on the 1W1B beamline at the Beijing Synchrotron Radiation Facility (BSRF). The electron beam energy of the storage ring was 2.5 GeV with a maximum current of 200 mA. The beamline energy was adjusted continuously by rotating a Si(111) double-crystal monochromator. The higher harmonics were rejected by fine-tuning the angle of the Si double crystal. At the sample position along the beamline, the energy resolution was $<1-3 \times 10^{-4}$ @ 9 keV, the luminous flux was $>4 \times 10^{11}$ photons/s @ 9 keV, and the beam size was 0.9 mm (H) $\times$ 0.3 mm (V). The X-ray absorption spectra were measured in transmission mode. The ionization chambers were used to detect the incident and transmitted beam signals. An Fe foil reference was used to calibrate the energy shift. The paraffin-protected samples were diluted with Vaseline and pressed to a self-supporting tablet for the test. The spectra were recorded in the energy range of 6900–7900 eV, including both the XANES and EXAFS regions. The average acquisition time for a spectrum was 16 min.

The X-ray fine structure (XAFS) spectra were processed by the Athena program in the IFEFFIT data analysis package.³⁷ The pre-edge region was fitted to a straight line by a Victoreen function, and the postedge was fitted with a cubic spline. The

background was removed by extrapolating the pre-edge line onto the EXAFS region, and the $\chi(E)$ data were normalized with respect to the edge jump step using the Athena program. The normalized $\chi(E)$ was transformed from energy space to k -space with $\chi(k)$ multiplied by k^2 to compensate for damping of EXAFS oscillations in the high- k region. Due to the significant noise in the EXAFS data at the higher k range, the k range was set to 2.5–10 Å. Subsequently, the k^2 -weighted $\chi(k)$ data were Fourier-transformed to R space. The compositions were fitted from XANES by using principle combination analysis (PCA) and the linear combination fitting (LCF) method. The processed $\chi(k)$ data were fitted in an R space ranging from 1.3 to 3.1 Å using the Artemis program. The structural fitting parameters used in the model included two lattice expansion parameters, one for iron oxide in the form of α -Fe₂O₃ and another one for iron carbides; three energy shift parameters, one for the first Fe–O shell of iron oxide, one for the remaining shells of iron oxide, and one for shells of iron carbides; three mean-square disorder parameters, one for the first Fe–O shell of iron oxide, one for the remaining shells of iron oxide, and one for shells of iron carbides; and the amplitude coefficients that account for the fraction of oxide and carbide, x for iron oxide and $1 - x$ for iron carbide. Structural parameters including coordination number (N), coordination distance (R), XAFS Debye–Waller factors, inner potential correction (ΔE_0), and estimated phase fractions were obtained. For all fits, the number of fitted parameters was lower than the number of independent parameters.

2.3. Catalyst Evaluation. The FTS experiments were conducted in a 12 mm (i.d.) stainless steel fixed-bed reactor with an isothermal bed length of 6 cm. Catalyst (40–60 mesh, 1 g) diluted with 3 g of quartz granule was loaded into the reactor; the remaining volume of the reactor was filled with quartz granules (20–40 and 40–60 mesh). A detailed description of the reactor and the product analysis systems has been given elsewhere.³⁸ The flow rates of the gases were controlled by mass flow meters separately (5850E, Brooks). The flow rate of the tail gas was measured by a wet-gas flow meter. Different iron carbides were prepared by using different atmospheres (CO, H₂, or 2H₂/CO). The reactor was first heated to 120 °C at a heating rate of 1.5 °C/min under a gas hourly space velocity (GHSV) of 4 NL/(g_{cat}·h) and maintained for 1 h to remove the adsorbed water. Then, the reactor was heated to 300 °C at a heating rate of 1 °C/min and maintained for 20 h. After treatment, the reactor was cooled to 150 °C. Then, the reactor was regulated to 3.0 MPa with 4 NL/(g_{cat}·h) of 2H₂/CO. The temperature was gradually increased to 260 °C at 10 °C/h. The tail gas was analyzed online by gas chromatographs. The nongaseous products in the hot and cold traps were collected over 24 h for calculation of the mass balance and sampled for analysis.

3. RESULTS AND DISCUSSION

3.1. FTS Performance and Iron Phase Structures of Catalysts. The elemental analysis of fresh calcined catalyst was characterized by XRF (Table S1) and ICP-AES (Table S2). Elements other than iron and silicon have a very low content and can be neglected. The iron content is consistent with the designed value of 50.9 wt %. Because the aim of this work is to study the relationship between iron carbide structures and their FTS performances, it is very crucial to evaluate the FTS reaction on a catalyst with stable active phase compositions. Therefore, the pretreatment strategy was carefully selected by

repeated experiments in the fixed-bed reactors in order to find a moderate reduction/carburization degree because insufficient carburization would lead to a long induction period while the excess carburization would result in rapid deactivation during FTS reaction.

After different pretreatments, FTS reaction was performed over the catalysts. The Fischer–Tropsch performances of catalysts are shown in Figure 1 and Table 1. The FeSi-CO

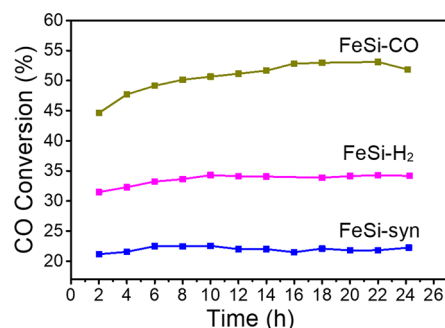


Figure 1. CO conversion of the catalysts after different treatments.

catalyst has the highest CO conversion (50.8%), and the FeSi-syn catalyst has the lowest conversion (22.3%). The iron time yields (FTYs) for FeSi-CO, FeSi-H₂, and FeSi-syn catalysts are 16.5, 10.8, and 7.2 $\mu\text{mol}_{\text{CO}}/\text{g}_{\text{Fe}}/\text{s}$, respectively. Suo et al.³³ treated 100Fe₂Si catalyst with 2H₂/CO at 300 °C for 20 h, and the FTY was 7.5 $\mu\text{mol}_{\text{CO}}/\text{g}_{\text{Fe}}/\text{s}$. This value is in agreement with that of the FeSi-syn catalyst in this work, indicating that it is the case for the activity of syngas-treated catalyst. CO or H₂ pretreatment results in different catalyst structures and leads to higher activities than FeSi-syn catalyst. Also, the three catalysts present distinct product selectivity, indicating that the active sites in different catalysts during FTS reaction are not identical.¹³ In order to obtain a comparative CO conversion with FeSi-CO catalyst, a prolonged reduction time and lower reaction space velocity were applied to FeSi-H₂ and FeSi-syn catalysts. It is found that the hydrocarbon selectivity is almost not affected by the CO conversion level. FeSi-CO catalyst has the highest CH₄ selectivity and lowest C₅₊ selectivity. Surprisingly, FeSi-H₂ catalyst presents the lowest CH₄ selectivity (10.6%) and highest C₅₊ selectivity (74.4%), which is extraordinary among promoter-free Fe/SiO₂ catalysts. For example, Suo et al.³³ reported 15.6% CH₄ selectivity in an Fe/SiO₂ catalyst mainly containing γ -Fe₃C₂ carbide using 2H₂/CO treatment; Pérez De Berti et al.³⁹ reported ~18 or ~31% CH₄

selectivity on Fe/SBA-15 catalyst containing 22 or 14% carbides (χ -Fe₃C₂ + ϵ -Fe_{2.2}C). In addition, the prolonged reduction time increases the FTY due to the increased active phase content, while the variation of space velocity does not change the FTY value.

The calcined, reduced, and spent catalysts were characterized to figure out the iron-based phases in these catalysts. The XRD pattern of calcined catalyst (Figure 2) shows two broad peaks

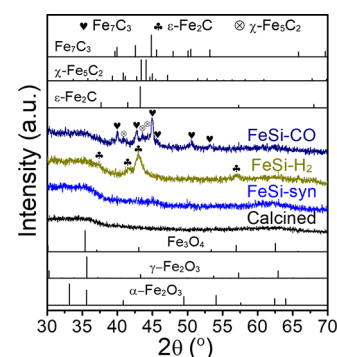


Figure 2. XRD patterns of the calcined and spent catalysts.

located at 34.7 and 62.7°, indicating that the sample is highly dispersed. The iron phase might be α -Fe₂O₃ (JCPDS 33-0664) or γ -Fe₂O₃ (JCPDS 39-1346) according to the ICDD database. The XRD, Fourier-transformed EXAFS, and MES results of reduced catalysts are shown in Figures S1–S3 and Tables S3 and S4. The data has proved that the distinct Fe₇C₃ phase is formed in the reduced FeSi-CO catalyst. α -Fe is formed in the reduced FeSi-H₂ catalyst. A broad weak peak at 42–46° is attributed to iron carbide observed in reduced FeSi-syn catalyst. Because the catalysts were carburized to a moderate degree, the broad peaks corresponding to unreduced iron oxide still can be observed. The detailed phase structures, particle sizes, and compositions of reduced catalysts are given in Table 2. Because the iron phase structures in working catalysts are directly related to FTS performance, the spent catalysts were withdrawn safely after reaching the steady-state reactions. The structure, content, and size of each iron phase in these spent catalysts were identified by XRD, XAFS, MES, and TEM.

Figure 2 shows the XRD spectra of spent catalysts. The spent FeSi-CO catalyst displays diffraction peaks at 40.0, 42.9, 44.9, 50.8, and 53.4°, which are attributed to hcp Fe₇C₃ (JCPDS 17-0333). The estimated crystallite size of Fe₇C₃ using the peak at 44.9° is about 28.2 nm. The spent FeSi-H₂ catalyst displays diffraction peaks at 37.7, 41.4, 43.2, and 57.3°, which are

Table 1. Activity and Hydrocarbon Selectivity of Different Catalysts^a

	CO conv. (%)	CO ₂ selec. (C mol %)	hydrocarbon selec. (wt %)			C _{2–4} °/C _{2–4}	FTY ^c ($\mu\text{mol}_{\text{CO}}/\text{g}_{\text{Fe}}/\text{s}$)
			CH ₄	C _{2–4}	C ₅₊		
FeSi-CO	50.8 ^a	9.7	19.2	30.8	50.0	1.00	16.5
FeSi-H ₂	33.1 ^a	4.6	10.6	15.0	74.4	0.78	10.8
	44.8 ^b	5.7	9.3	14.3	76.4	0.75	10.9
FeSi-syn	22.3 ^a	2.8	12.9	17.1	70.0	0.91	7.2
	32.3 ^c	3.6	12.5	20.1	67.4	0.83	7.9
	49.2 ^d	4.8	11.8	20.9	67.3	0.80	8.0

^aTreatment: 300 °C, 0.2 MPa, GHSV = 4000 h^{−1}, 20 h; FTS: 260 °C, 3 MPa, H₂/CO = 2, GHSV = 4000 h^{−1}. ^bTreatment: 300 °C, 0.2 MPa, GHSV = 4000 h^{−1}, 20 h; FTS: 260 °C, 3 MPa, H₂/CO = 2, GHSV = 3000 h^{−1}. ^cTreatment: 300 °C, 0.2 MPa, GHSV = 4000 h^{−1}, 30 h; FTS: 260 °C, 3 MPa, H₂/CO = 2, GHSV = 3000 h^{−1}. ^dTreatment: 300 °C, 0.2 MPa, GHSV = 4000 h^{−1}, 30 h; FTS: 260 °C, 3 MPa, H₂/CO = 2, GHSV = 2000 h^{−1}. ^eFTY: amount of consumed CO per gram of iron per second.

Table 2. Overview of the Physicochemical Properties of the Reduced and Spent Catalysts

catalysts		XRD	TEM	MES	XANES	EXAFS
calcined	—	Fe ₂ O ₃	α -Fe ₂ O ₃ (5.5 nm)	—	Fe ₂ O ₃	Fe ₂ O ₃
FeSi-H ₂	reduced	Fe ₃ O ₄	—	Fe ₃ O ₄ (74.6%)	—	Fe _x O _y
		α -Fe (15.4 nm)	—	Fe ²⁺ + α -Fe (25.4%)	—	α -Fe
	spent	Fe ₃ O ₄	Fe _x O _y	Fe ₃ O ₄ (74.3%)	Fe ₂ O ₃ (39.2%)	Fe _x O _y
		ϵ -Fe ₂ C (9.1 nm)	ϵ -Fe ₂ C (13.9 nm)	ϵ -Fe ₂ C (11.0%)	Fe ₃ O ₄ (35.6%)	ϵ -Fe ₂ C
FeSi-CO	reduced	Fe ₃ O ₄	—	χ -Fe ₅ C ₂ (14.7%)	Fe _x C _y (25.1%)	χ -Fe ₅ C ₂
		Fe ₇ C ₃ (26.0 nm)	—	Fe ₃ O ₄ (64.9%)	—	Fe _x O _y
		—	—	Fe ₇ C ₃ (15.4%)	—	Fe ₇ C ₃
		—	—	χ -Fe ₅ C ₂ (19.7%)	—	χ -Fe ₅ C ₂
	spent	Fe ₃ O ₄	Fe _x O _y	Fe ₃ O ₄ (67.2%)	Fe ₂ O ₃ (36.6%)	Fe _x O _y
		Fe ₇ C ₃ (28.2 nm)	Fe ₇ C ₃ (27.5 nm)	Fe ₇ C ₃ (16.3%)	Fe ₃ O ₄ (26.2%)	Fe ₇ C ₃
FeSi-syn	reduced	Fe ₃ O ₄	—	Fe ₃ O ₄ (83.9%)	—	Fe _x O _y
		χ -Fe ₅ C ₂	—	χ -Fe ₅ C ₂ (16.1%)	—	χ -Fe ₅ C ₂
	spent	Fe ₃ O ₄	Fe _x O _y	Fe ₃ O ₄ (86.9%)	Fe ₂ O ₃ (53.5%)	Fe _x O _y
		χ -Fe ₅ C ₂	χ -Fe ₅ C ₂ (7.5 nm)	χ -Fe ₅ C ₂ (13.1%)	Fe ₃ O ₄ (25.3%)	χ -Fe ₅ C ₂
FeSi-CO-H ₂	spent	Fe ₃ O ₄	Fe _x O _y	Fe ₃ O ₄ (62.0%)	—	—
		ϵ -Fe ₂ C (10.6 nm)	ϵ -Fe ₂ C (15.4 nm)	ϵ -Fe ₂ C (22.7%)	Fe _x C _y (21.2%)	—
		—	χ -Fe ₅ C ₂ (7.8 nm)	χ -Fe ₅ C ₂ (15.3%)	—	—

attributed to ϵ -Fe₂C (ϵ -Fe₂C or ϵ' -Fe_{2.2}C structures: JCPDS 36-1249). Because of the structural similarity of ϵ -Fe₂C or ϵ' -Fe_{2.2}C, ϵ -Fe₂C is tentatively assigned to represent this phase. It is indicated that the formed α -Fe evolved into ϵ -Fe₂C during FTS. The particle size of ϵ -Fe₂C based on the peak at 42.8° is about 9.1 nm. However, the broad contribution underlying the sharp peaks of Fe₇C₃ and ϵ -Fe₂C suggests the existence of some other poorly crystalline or smaller iron carbides. Three lines at 41.0, 43.6 and 44.7° in the spent FeSi-CO catalyst might originate from χ -Fe₅C₂. These unresolved carbides are beyond the detection limits of XRD and need to be identified by other complementary techniques. The XRD pattern of the spent FeSi-syn catalyst is similar to that of reduced FeSi-syn catalyst. The broad weak swell at 42–46° is characteristic of iron carbide. Because of the overlap of characteristic peaks of Fe₇C₃, ϵ -Fe₂C, and χ -Fe₅C₂ in these 2 θ degrees, the precise phase structures and compositions in the spent catalysts cannot be identified by the XRD method.

The nonphase corrected, k^2 weighted, Fourier transformed EXAFS spectra and fitted parameters for the spent catalysts are shown in Figure 3 and Table 3. The calcined catalyst shows three distinct scattering peaks (1.44, 2.61, and 3.26 Å), characteristics of α -Fe₂O₃ or γ -Fe₂O₃. The peak at 1.44 Å corresponds to the first shell of Fe–O, and the peak at 2.61 Å corresponds to the second shell of Fe–Fe. The much lower contribution at the second shell than that at the first shell, relative to standards, suggests that iron oxide particles are small. This result together with the XRD pattern confirms that the calcined catalyst is highly dispersed. As for spent catalysts, the contributions from 1.44 and 2.61 Å are attributed to unreduced iron oxide. However, the different shape and broadening of the shell at 1.44 Å indicate that different iron oxide and iron carbide species coexist as both Fe–C and Fe–O bonds affect this peak. Noteworthy differences are observed in other peaks for the spent catalysts. For the FeSi-CO catalyst, a shoulder at 1.80–2.13 Å alongside the first peak is obviously found, which is attributed to the Fe–Fe shell of iron carbides. In order to identify the contribution from different carbides, the EXAFS

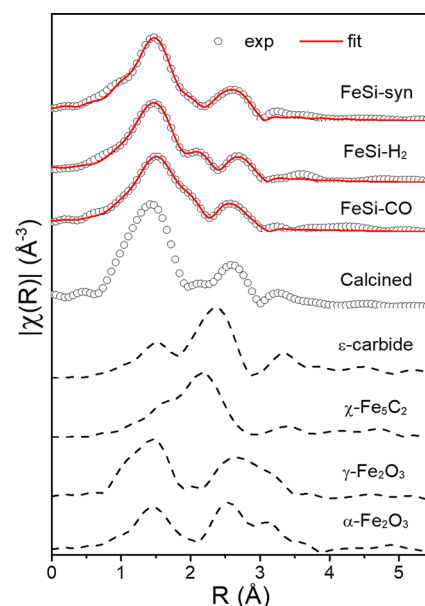


Figure 3. Fourier-transformed EXAFS spectra of the spent catalysts.

spectra of carbide standards are performed. ϵ -Fe₂C and χ -Fe₅C₂ standard carbides were synthesized according to the literature,^{25,28} and the phases were confirmed by XRD and MES (Figure S4). EXAFS of Fe₇C₃ is calculated by FEFF using cell structure information. The peak center of the Fe–Fe shell of χ -Fe₅C₂ is 2.17 Å, and for Fe₇C₃, it is 2.06 Å. The shoulder in the spent FeSi-CO catalyst is contributed by χ -Fe₅C₂ or Fe₇C₃. Quantitative fitting analysis of EXAFS is carried out to acquire the coordination numbers (CNs), as well as rough compositions of iron oxide and iron carbide. The results demonstrate that the spent FeSi-CO catalyst has 43% iron carbide and 57% iron oxide. The CNs of Fe–C and Fe–Fe in the iron carbide are 1.7 and 3.4, respectively. The CNs of Fe–O and Fe–Fe in the iron oxide are 5.6 and 3.3, respectively. For the spent FeSi-H₂ catalyst, a new detached peak at 2.11 Å

Table 3. Results of EXAFS Fitting for the Spent Catalysts

spent catalysts	iron phases	compositions	shell	N	R (Å)	σ^2 (Å ²)	ΔE (eV)
FeSi-H ₂	iron oxide	0.56	Fe–O	3.1	1.94	0.0098	–1.6
			Fe–Fe	4.7	3.05	0.016	4.0
	iron carbide	0.44	Fe–C	3.7	2.04	0.020	6.3
			Fe–Fe	4.9	2.75	0.020	6.3
FeSi-CO	iron oxide	0.57	Fe–O	5.6	1.94–2.12	0.0086	8.0
			Fe–Fe	3.3	2.97	0.019	3.1
	iron carbide	0.43	Fe–C	1.7	2.00	0.0080	–11.5
			Fe–Fe	3.4	2.39–2.62	0.0080	–11.5
FeSi-syn	iron oxide	0.77	Fe–O	4.1	1.95	0.010	1.1
			Fe–Fe	6.1	3.06	0.020	8.9
	iron carbide	0.23	Fe–C	1.2	2.24	0.020	11.0
			Fe–Fe	3.1	2.63–2.86	0.020	11.0

appears, which is due to ϵ -Fe₂C. The Fe–Fe bond lengths in χ -Fe₅C₂ and Fe₇C₃ are 2.17 and 2.06 Å, respectively, and that bond length in ϵ -Fe₂C is a little longer, reaching 2.33 Å. Due to this longer Fe–Fe coordination in ϵ -Fe₂C, the contribution at 2.11 Å in the spent FeSi-H₂ catalyst is detached to form a new peak. Furthermore, the fitted CN in the iron carbide part is the highest among all of the spent catalysts. This can also help to confirm the existence of ϵ -Fe₂C because of the highest carbon content in ϵ -Fe₂C. For the spent FeSi-syn catalyst, the EXAFS spectrum is somewhat the same as the calcined catalyst, verifying that 2H₂/CO had the weakest reducing ability. The broadening contributions at 1.82 and 2.35 Å are derived from χ -Fe₅C₂.

The spent catalysts after FTS were characterized by MES at 11 K because MES specializes in determining the phase structure and content of iron carbides. The spectra and fitted results are shown in Figure S5 and Table 4. It can be seen that all spectra are the result of the superposition of sextet subspectra. No curved background appeared in the spectra,

indicating that the iron oxides and iron carbides in the catalysts are fully magnetically blocked at 11 K. No doublet of paramagnetic Fe³⁺ or Fe²⁺ is found, meaning that no Fe³⁺ or Fe²⁺ diffuses inside of SiO₂. The spent FeSi-syn catalyst spectrum is fitted with five sextets. The two sextets with Hhf values of 490 and 440 kOe are attributed to the tetrahedral and octahedral sites in Fe₃O₄,⁴⁰ while the remaining three sextets with Hhf values of 235, 202, and 110 kOe are attributed to the three crystallographic sites in χ -Fe₅C₂.^{41–43} The composition of χ -Fe₅C₂ in spent FeSi-syn catalyst is 13.1%. MES results can further confirm that the weak, broad carbide swell in XRD of the spent FeSi-syn catalyst (Figure 2) is from χ -Fe₅C₂. The fitting procedures for spent FeSi-H₂ catalyst are the same as those in the spent FeSi-syn catalyst. The spectrum is fitted with two sextets for the two Fe₃O₄ subspectra and three sextets (175, 243, and 120 kOe) for ϵ -Fe₂C.⁴⁴ It is necessary to add sextets of χ -Fe₅C₂ to obtain a satisfactory fitting, which indicates that the unsolved species in XRD is χ -Fe₅C₂. The compositions of χ -Fe₅C₂ and ϵ -Fe₂C are 11.0 and 14.7%, respectively. Fe₃O₄ and Fe₇C₃ (160, 182, 208, and 235 kOe)⁴⁵ are identified in the spent FeSi-CO catalyst. Likewise, sextets of χ -Fe₅C₂ are needed to converge the fit to a minimum, which suggests the existence of χ -Fe₅C₂ in the catalyst. The final fitting gives the composition of 16.5% χ -Fe₅C₂ and 16.3% Fe₇C₃. The contents of total iron carbides in the spent FeSi-CO, FeSi-H₂, and FeSi-syn catalysts are 32.8, 25.7, and 13.1%, respectively. Compared with that in the reduced catalysts (Table 2, Figure S3, and Table S4), the contents of Fe⁰ in catalysts remain consistent in reaction, suggesting that the catalysts are stable during FTS.

It is accepted that the reduction degree and the content of iron carbides play important roles in determining the activity of the catalyst by providing different quantities of active sites. XANES of spent catalysts was performed to confirm the contents from MES. Figure 4 shows XANES spectra and LCF results of spent catalysts. The XANES spectra of catalysts are located in between the α -Fe reference sample and calcined catalyst, indicating that the catalysts are composed of different iron oxides and Fe⁰ (i.e., carbides). From the representative features (the locations of the pre-edge and absorption edge) of XANES, the reduction degree of spent FeSi-CO catalyst is higher than that of the spent FeSi-H₂ catalyst, followed by the spent FeSi-syn catalyst. The fitted phase compositions have confirmed this observation. The spent FeSi-CO catalyst has the most iron carbide in all catalysts, and the FeSi-syn catalyst has the least. The contents of iron carbides fitted from XANES are consistent with the MES results.

Table 4. Iron Phase Compositions of the Spent Catalysts from MES at 11 K

spent catalysts	IS (mm/s)	Hhf (kOe)	area (%)	phase	compositions (%)
FeSi-H ₂	0.47	490	41.0	Fe ₃ O ₄ (A)	74.3
	0.52	440	33.3	Fe ₃ O ₄ (B)	
	0.40	233	3.0	χ -Fe ₅ C ₂ (I)	
	0.38	190	5.0	χ -Fe ₅ C ₂ (II)	11.0
	0.34	110	3.0	χ -Fe ₅ C ₂ (III)	
	0.40	175	4.1	ϵ -Fe ₂ C (I)	14.7
	0.20	243	6.7	ϵ -Fe ₂ C (II)	
	0.00	120	3.9	ϵ -Fe ₂ C (III)	
FeSi-CO	0.40	490	33.7	Fe ₃ O ₄ (A)	67.2
	0.65	440	33.5	Fe ₃ O ₄ (B)	
	0.40	235	6.4	χ -Fe ₅ C ₂ (I)	16.5
	0.36	202	4.1	χ -Fe ₅ C ₂ (II)	
	0.40	135	6.0	χ -Fe ₅ C ₂ (III)	
	0.38	160	3.4	Fe ₇ C ₃ (I)	16.3
	0.40	182	5.8	Fe ₇ C ₃ (II)	
	0.16	208	3.5	Fe ₇ C ₃ (III)	
	0.12	235	3.6	Fe ₇ C ₃ (IV)	13.1
	0.49	490	49.2	Fe ₃ O ₄ (A)	
FeSi-syn	0.50	440	37.7	Fe ₃ O ₄ (B)	
	0.38	235	5.8	χ -Fe ₅ C ₂ (I)	13.1
	0.40	202	3.0	χ -Fe ₅ C ₂ (II)	
	0.40	110	4.3	χ -Fe ₅ C ₂ (III)	

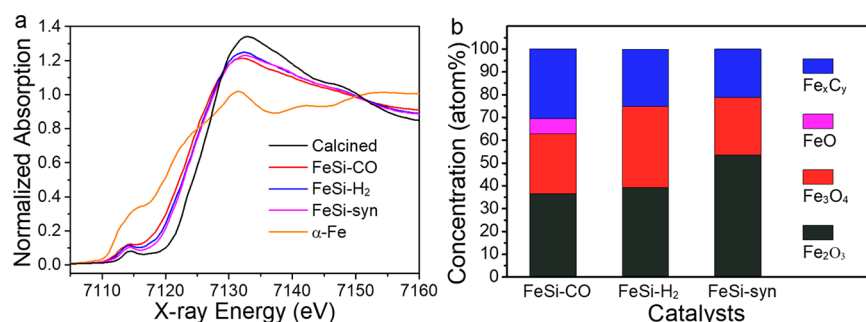


Figure 4. (a) XANES spectra and (b) fitted compositions of the spent catalysts.

The phase structure and particle size distribution (PSD) of spent catalysts were characterized by (HR)TEM, shown in Figure 5. In the calcined catalyst (Figure 5a), the iron oxide

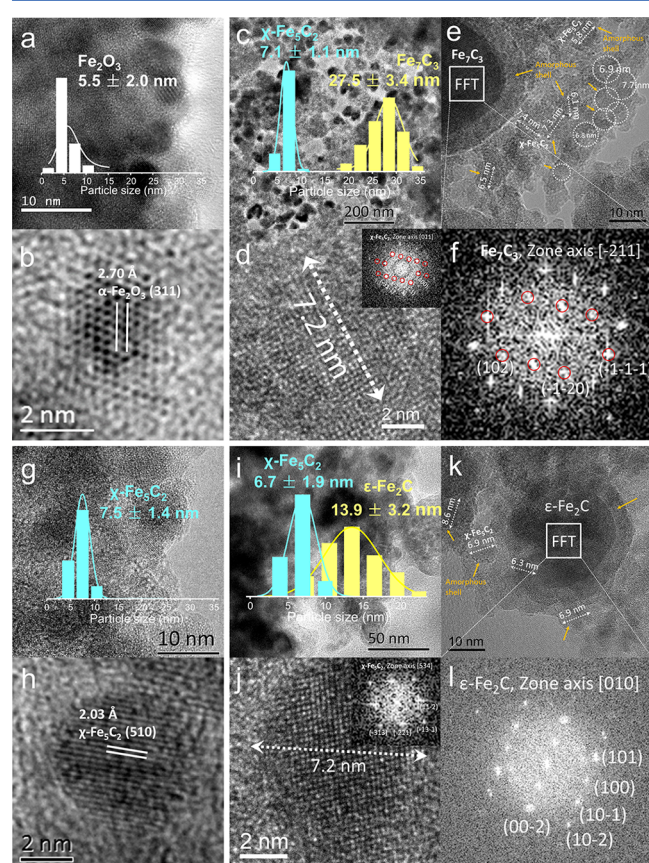


Figure 5. TEM images of the (a,b) calcined, (c–f) spent FeSi-CO, (g,h) spent FeSi-syn, and (i–l) spent FeSi-H₂ catalysts. (f,l) FFT images of the corresponding selected area with the plot marked FFT in (e) and (k). The PSD of each catalyst is inset in (a), (c), (g), and (i). Insets in (d) and (j) are FFT images of the HRTEM images.

nanoparticles are finely distributed and stabilized by SiO₂. The average statistical size from TEM is 5.5 nm. The interplanar distance of 2.70 Å in HRTEM (Figure 5b) corresponds to the (311) plane of α -Fe₂O₃. The spent FeSi-CO catalyst (Figure 5c–f) shows many large particles besides the small ones. The STEM elementary mapping (Figure S6) suggests that the particles are iron carbides. An amorphous shell is also observed, which might be iron oxides resulting from the mild surface oxidation during sample preparation and the TEM test.^{46–48} The coated paraffin was removed by sonication in ethanol,

which slightly impaired the protecting effect. Different carbide phases can be identified by FFT images and the interplanar distance of lattice fringes. The large particle in Figure 5e,f is identified as Fe₇C₃ along zone axis [−211] by comparing the FFT image with the simulated single-crystal diffraction pattern. The average particle size of Fe₇C₃ is about 27.5 nm, which is close to the estimated size from XRD (28.2 nm). The FFT image of the small particle (Figure 5d) is indexed to χ -Fe₅C₂ along with zone axis [011], and the lattice fringes throughout the whole particle manifest a single crystal. Carbide can be further confirmed by the surrounding amorphous shell due to the mild surface oxidation of carbides during the TEM test. Those small χ -Fe₅C₂ particles are marked with dashed circles or double arrow lines, as shown in Figure 5e. The yellow arrows indicate the amorphous shell. The sizes of partial particles are labeled for comparison. The average particle size of χ -Fe₅C₂ is 7.1 nm, in agreement with the conclusion from XRD that χ -Fe₅C₂ existed in small particles. More evidence about phase and size identification in this catalyst is presented in Figure S7. In the spent FeSi-syn catalyst (Figure 5g,h), no obvious particle agglomeration is found, and the catalyst is still dispersed well, similar to the calcined catalyst. Figure 5h is a HRTEM image of particles with sizes around 7 nm. Its lattice fringe of 2.03 Å corresponds to the (510) plane of χ -Fe₅C₂. The statistic average particle size of χ -Fe₅C₂ is about 7.5 nm. For the spent FeSi-H₂ catalyst (Figure 5i–l), the sizes of particles mainly range from about 5 to 25 nm. The relatively large particle is demonstrated to be ϵ -Fe₂C along with zone axis [010] (Figure 5k,l), and the relatively small particle is attributed to χ -Fe₅C₂ along [534] (Figure 5j). For some carbide particles that are difficult to acquire high-quality images, several spots are used to distinguish their most probable phase (Figure S8a inset). The particle sizes of both carbides are analyzed via a statistical approach. The average sizes of ϵ -Fe₂C and χ -Fe₅C₂ are 13.9 and 6.7 nm, respectively. More evidence about this catalyst analysis is presented in Figure S8. The size of ϵ -Fe₂C is larger than that from XRD (9.1 nm), which is due to the fact that the peak used to calculate the size of ϵ -Fe₂C comprised the contribution of χ -Fe₅C₂ (leading to a higher value of full width at half-maximum). It has been reported that different sized carbide particles are formed in the working catalyst.¹⁹ Our results clearly demonstrate a bimodal nanoparticle distribution in the spent FeSi-CO and FeSi-H₂ catalysts. The χ -Fe₅C₂ tends to be small, while Fe₇C₃ and ϵ -Fe₂C are large.

The present study indicates that Fe₇C₃ can be formed and exist stably at typical FTS temperatures between 260 and 300 °C. It can provide a great convenience to study the structures and FTS performances of Fe₇C₃ and avoid being affected by carbon deposition, catalyst attrition, element loss, and poison-

ing in long reaction runs. In order to confirm the formation of Fe_7C_3 under MTFT conditions, in situ MES of the catalyst was performed at more moderate treatment conditions (CO , 1 bar, 280°C , and 20 h), and the results are shown in Figure 6 and

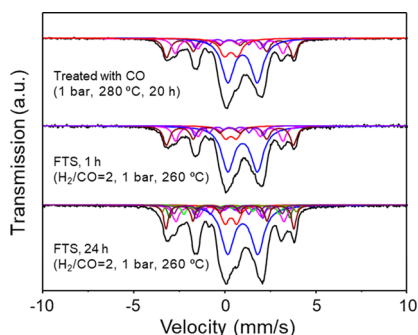


Figure 6. In situ MES spectra of CO-treated catalyst and spent catalysts after FTS reaction for 1 and 24 h measured at 300 K.

Table 5. The results prove that the treated FeSi-CO catalyst contains 58% iron oxide (spm Fe^{3+} and Fe^{2+}), 26% $\chi\text{-Fe}_5\text{C}_2$, and 16% Fe_7C_3 . The content of Fe_7C_3 in the in situ treated FeSi-CO catalyst is close to the value of 16.3% in the ex situ reduced FeSi-CO catalyst. After carburization, the catalyst was exposed to syngas atmosphere ($2\text{H}_2/\text{CO}$, 260°C , 1 bar). From the MES, it is clear that the iron phase compositions are nearly unchanged after 1 h. After 24 h, the content of iron oxide decreases to 45%, indicating that further carburization takes place during the reaction. The content of $\chi\text{-Fe}_5\text{C}_2$ increases to 35%, while the content of Fe_7C_3 slightly increases to 20%. The further carburization is probably due to the setup discrepancy

and the quite different reaction atmospheres compared with the fixed-bed reactor. In the in situ MES measurement, a very small amount of catalyst was loaded to ensure sufficient transmission of the γ -ray. This led to a very high gas hourly space velocity, relatively low syngas conversion, and very low H_2O or CO_2 partial pressures (imposing high μ_{C} and low μ_{O}), which means that the μ_{C} in the MES cell is relatively higher than that in fixed-bed reactor. Therefore, the sample is further carburized in the MES reactor. Because of these differences, the in situ MES results are used to testify the formation of Fe_7C_3 under medium-temperature FTS conditions but not to explore the structure–performance relationship of different iron carbides.

Because the optimum temperature for $\chi\text{-Fe}_5\text{C}_2$ formation is $250\text{--}350^\circ\text{C}$, $\chi\text{-Fe}_5\text{C}_2$ is frequently observed in FTS carried out on this temperature scale.^{11,14,24,49} The direct conversion of iron oxide to $\epsilon\text{-Fe}_2\text{C}$ is kinetically and entropically unfavorable, and $\epsilon\text{-Fe}_2\text{C}$ is often synthesized by carburization of $\alpha\text{-Fe}$.^{12,50} Therefore, it is regular and reasonable to observe $\epsilon\text{-Fe}_2\text{C}$ only in the spent FeSi- H_2 catalyst and $\chi\text{-Fe}_5\text{C}_2$ in all spent catalysts. However, it is rare to observe Fe_7C_3 in the iron-based catalysts during pretreatment or FTS reaction. Only a few research reports have reported this phase.^{16–19} Eckstrom and Adcock¹⁶ first observed the Fe_7C_3 phase along with Fe_3O_4 and Hägg carbide during the course of a fluidized FTS run at 2.7 MPa and 360°C . Lately, Datye et al.¹⁹ and Mansker et al.¹⁸ observed the Fe_7C_3 phase in CO-pretreated commercial Fe/SiO₂ catalyst after long FTS runs. Clearly, the formation of Fe_7C_3 is dependent on a specific carburization atmosphere and is also affected by kinetic factors (e.g., long reaction time). de Smit et al.²⁴ applied ab initio atomistic thermodynamics to study the stability of iron carbides. They found that Fe_7C_3 is stable with

Table 5. Iron Phase Compositions of in Situ CO-Treated Catalyst and Spent Catalysts after in Situ FTS Reaction for 1 and 24 h

catalysts	IS ($\text{mm}\cdot\text{s}^{-1}$)	QS ($\text{mm}\cdot\text{s}^{-1}$)	Hhf (kOe)	Γ ($\text{mm}\cdot\text{s}^{-1}$)	area (%)	phase	compositions (%)
reduced with CO (1 bar, 280°C , 20 h)	0.34	0.64		0.5	11	spm Fe^{3+}	58
	0.97	1.62		0.85	47	Fe^{2+}	
	0.29		218	0.29	11	$\chi\text{-Fe}_5\text{C}_2$ (I)	26
	0.28		184	0.44	9	$\chi\text{-Fe}_5\text{C}_2$ (II)	
	0.26		114	0.31	6	$\chi\text{-Fe}_5\text{C}_2$ (III)	
	0.2	0.34	163	0.3	3	Fe_7C_3 (I)	16
	0.21	−0.05	185	0.3	5	Fe_7C_3 (II)	
	0.25	0.04	205	0.25	5	Fe_7C_3 (III)	
	0.32	−0.16	229	0.3	3	Fe_7C_3 (IV)	
FTS, 1 h ($\text{H}_2/\text{CO} = 2$, 1 bar, 260°C)	0.34	0.57		0.5	8	spm Fe^{3+}	55
	0.98	1.63		0.83	47	Fe^{2+}	
	0.29		218	0.29	12	$\chi\text{-Fe}_5\text{C}_2$ (I)	28
	0.28		184	0.44	10	$\chi\text{-Fe}_5\text{C}_2$ (II)	
	0.26		114	0.31	6	$\chi\text{-Fe}_5\text{C}_2$ (III)	
	0.2	0.34	163	0.3	4	Fe_7C_3 (I)	17
	0.21	−0.05	185	0.3	5	Fe_7C_3 (II)	
	0.25	0.04	205	0.25	5	Fe_7C_3 (III)	
	0.32	−0.16	229	0.3	3	Fe_7C_3 (IV)	
FTS, 24 h ($\text{H}_2/\text{CO} = 2$, 1 bar, 260°C)	0.34	0.64		0.5	8	spm Fe^{3+}	45
	0.99	1.65		0.83	37	Fe^{2+}	
	0.29		218	0.29	14	$\chi\text{-Fe}_5\text{C}_2$ (I)	35
	0.28		184	0.44	14	$\chi\text{-Fe}_5\text{C}_2$ (II)	
	0.26		114	0.31	7	$\chi\text{-Fe}_5\text{C}_2$ (III)	
	0.2	0.34	163	0.3	5	Fe_7C_3 (I)	20
	0.21	−0.05	185	0.3	5	Fe_7C_3 (II)	
	0.25	0.04	205	0.25	6	Fe_7C_3 (III)	
	0.32	−0.16	229	0.3	4	Fe_7C_3 (IV)	

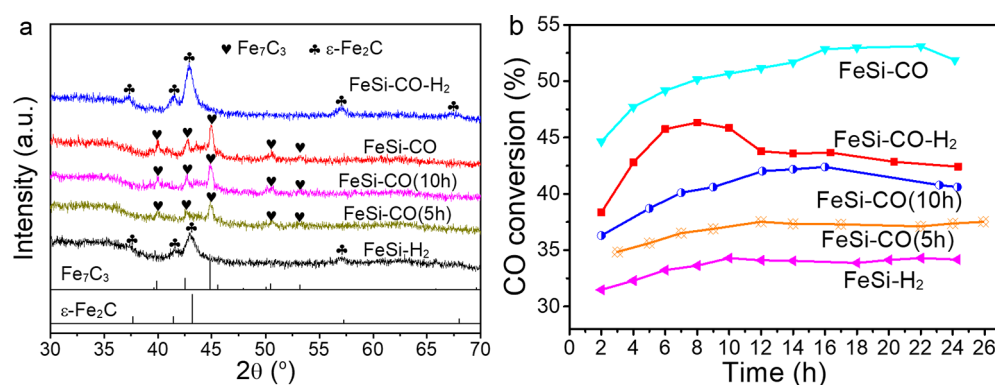


Figure 7. (a) XRD spectra and (b) CO conversion of the catalysts in control experiments.

Table 6. Intrinsic Activity of ϵ -Fe₂C, Fe₇C₃, and χ -Fe₅C₂

spent catalysts	FTY ($\mu\text{mol}_{\text{CO}}/(\text{g}_{\text{Fe}}\text{s})$)	carbide	FTY contribution (%)	carbide content (%)	size (nm)	surface area ($\text{m}^2/\text{g}_{\text{Fe}}$)	number of active sites ($\times 10^{-4}$ mol/g _{Fe})	TOF ($\times 10^{-2}$ s ⁻¹)
FeSi-syn	7.2	χ -Fe ₅ C ₂	100	13.1	7.5	15.0	4.44	1.62
FeSi-H ₂	10.8	χ -Fe ₅ C ₂	63.0	11.0	6.7	14.1	4.18	1.62
		ϵ -Fe ₂ C	37.0	14.7	13.9	9.1	2.69	1.50
FeSi-CO	16.5	χ -Fe ₅ C ₂	58.2	16.5	7.1	20.0	5.91	1.62
		Fe ₇ C ₃	41.8	16.3	27.5	5.1	1.51	4.59
FeSi-CO-H ₂	13.8	χ -Fe ₅ C ₂	62.3	15.3	7.8	16.9	4.99	1.62
		ϵ -Fe ₂ C	37.7	22.7	15.4	12.7	3.75	1.52

respect to χ -Fe₅C₂ at higher μ_{C} (~ 7.25 eV). At low temperatures (imposing high μ_{C}), however, there is significant (kinetic) competition between the formation of the Fe₇C₃ phase and ϵ -carbides. This might explain the fact that the Fe₇C₃ phase is difficult to form experimentally because at low temperature there is competition with the ϵ -carbide phases and at higher temperature (low μ_{C}) conditions χ -Fe₅C₂ and θ -Fe₃C are more stable. In the present study, Fe₇C₃ is elaborately fabricated by treating highly dispersed ultrasmall iron oxide particles with pure CO at moderate temperature. The pure CO endows a high μ_{C} , and moderate temperature is used to mainly suppress the ϵ -Fe₂C. It seems that the small size kinetically favors the carbon diffusion leading to formation of Fe₇C₃. Besides, the silica support probably plays a vital role in the formation of Fe₇C₃. In those limited reports on the identification of Fe₇C₃, only silica-supported iron catalysts were reported to form Fe₇C₃ during FTS.^{18,19} Nevertheless, Fe₇C₃ does not completely win the competition with χ -Fe₅C₂ by using the present pretreatment strategy. The in-depth mechanism of Fe₇C₃ formation and the key influence factors need to be further studied.

3.2. Relationship between Active Phases and Catalytic Performances. It has been reported that the activity of a Fischer–Tropsch iron-based catalyst is influenced by reduction degrees, phases, and particle sizes of the active phases. The reduction degree is important for the catalytic activity as it could affect the amount of surface active sites in the working catalyst.^{46,51} In the present study, CO pretreatment produces the highest reduction degree on the catalyst, and 2H₂/CO syngas has the lowest reduction ability. The low reduction ability of syngas could be due to the complex reduction/carburization atmosphere in syngas pretreatment. During CO or H₂ pretreatment, only the reduction or carburization reaction of Fe₂O₃ takes place. The amount of byproduct (H₂O or CO₂) in the effluent is dependent on the sample

amount. However, for syngas pretreatment, once the catalyst is reduced or carburized, the accompanying FTS and WGS reactions would continuously consume syngas and decrease the partial pressures of H₂ and CO. The lower H₂ and CO partial pressures in the syngas pretreatment would severely decrease the carbon chemical potential μ_{C} and result in a low reduction/carburization degree compared with H₂ or CO pretreatment. It is found that the reduction degree order coincides with the FTS activity for the different pretreatments. The influence of reduction degree is also reflected by the activity order of FeSi-CO > FeSi-CO(10h) > FeSi-CO(5h) when reducing the calcined catalyst with CO for different times (Figure 7). However, reduction degree is not the only descriptor for the activity. For example, the spent FeSi-CO-H₂ catalyst has higher carbide content (38.0%, Figure S9 and Table S5) than the spent FeSi-CO catalyst (32.8%) but exhibits a lower activity. XRD results indicate that Fe₇C₃ carbide is contained in all of the spent FeSi-CO(5h), FeSi-CO(10h), and FeSi-CO catalysts, and ϵ -Fe₂C carbide is contained in the spent FeSi-H₂ and FeSi-CO-H₂ catalysts (Figure 7). These discrepancies suggest that different carbide phases have an important influence on FTS activity.

The effect of different iron carbide phases on FTS performance has been studied by many groups.^{13,18,19,23,51–55} However, the contribution of intrinsic activity from different carbides has been rarely reported.²⁶ It needs to be noted that it is not precise to correlate the activity only with the quantitative amount of carbides. To compare the intrinsic activity of iron carbides, both the carbide content and the carbide particle size should be taken into consideration as these two factors determine the number of active sites. It has been proved that the particle size can strongly affect the catalytic performance, and the critical size for FTS reaction was about 6 nm.^{56–64} The TOF and C₅₊ selectivity increase with the size up to the critical size and then remain almost unchanged with a further increase

in size. In our work, the particle sizes of χ -Fe₅C₂ in three catalysts are in the range of 6.7–7.5 nm, which is slightly higher than the critical size. Furthermore, the difference between the sizes of these χ -Fe₅C₂ in different catalysts is small. Therefore, it can be assumed that the χ -Fe₅C₂ in three catalysts has identical intrinsic activity.

Because the sole carbide in the spent FeSi-syn catalyst is χ -Fe₅C₂, this catalyst can serve as a benchmark to derive the activities of other carbides. The content of χ -Fe₅C₂ in the spent FeSi-syn catalyst is 13.1% (MES), and the particle size of χ -Fe₅C₂ is 7.5 nm (TEM). On the basis of these two values, the surface area of χ -Fe₅C₂ in the spent FeSi-syn catalyst is calculated to be 15.0 m²/g_{Fe}. Considering that the surface iron density for iron carbide is about 1.78×10^{19} atoms/m², the number of active sites is 4.44×10^{-4} mol/g_{Fe}. At last, the intrinsic activity (TOF) of χ -Fe₅C₂ is determined to be 1.62×10^{-2} s⁻¹. The calculated intrinsic activities of different iron carbides are presented in Table 6. For the spent FeSi-H₂ catalyst, both χ -Fe₅C₂ and ϵ -Fe₂C contribute to the overall FTS activity. Their respective surface areas and numbers of active sites are calculated based on their phase contents and particle sizes. The FTY contribution from χ -Fe₅C₂ in spent FeSi-H₂ catalyst is calculated as $6.8 \mu\text{mol}_{\text{CO}}/(\text{g}_{\text{Fe}}\cdot\text{s})$. This value is subtracted from the total FTY of FeSi-H₂ catalyst ($10.8 \mu\text{mol}_{\text{CO}}/(\text{g}_{\text{Fe}}\cdot\text{s})$). The resulting value of $4.0 \mu\text{mol}_{\text{CO}}/(\text{g}_{\text{Fe}}\cdot\text{s})$ is attributed to the FTY contribution from ϵ -Fe₂C. On the basis of the values of the FTY and active site amount, the TOF of ϵ -Fe₂C is calculated to be 1.50×10^{-2} s⁻¹. For the spent FeSi-CO catalyst, χ -Fe₅C₂ and Fe₇C₃ contribute to the overall FTS activity. Likewise, the TOF of Fe₇C₃ is calculated to be 4.59×10^{-2} s⁻¹. It can be found that Fe₇C₃ has the highest intrinsic activity among these three iron carbides. The intrinsic activity of Fe₇C₃ is three times as high as those of χ -Fe₅C₂ and ϵ -Fe₂C. The high CO conversion in FeSi-CO catalyst lies not only in the highest carbide content but also in the highest intrinsic activity of Fe₇C₃. The superior activity of Fe₇C₃ can also be verified by the result that FeSi-CO catalyst has a higher activity than FeSi-CO-H₂ catalyst. In order to testify the feasibility of the calculation method in deriving the TOFs of different carbides in the present work, the spent FeSi-CO-H₂ catalyst is analyzed again. The iron phase compositions are acquired from MES at 11 K: 15.3% χ -Fe₅C₂, 22.7% ϵ -Fe₂C, and 62.0% Fe₃O₄ (Figure S9 and Table S5). The structures of ϵ -Fe₂C and χ -Fe₅C₂ are analyzed by TEM (Figure 8). The size distribution shows an average particle size of 7.8 nm for χ -Fe₅C₂ and 15.4 nm for ϵ -Fe₂C. The same calculation strategy is used to derive the intrinsic activity of ϵ -Fe₂C, and it turns out to be 1.52×10^{-2} s⁻¹. This value is consistent with that from FeSi-H₂ catalyst, which proves the reliability of our calculation method.

We compare the TOFs in the present catalysts with results reported in the literature (Table S6). Suo et al.³³ prepared a series of Fe/ α -SiO₂ catalysts via a coprecipitation method. During 2H₂/CO reduction/carburization and subsequent FTS reactions (280 °C, 1.5 MPa, H₂/CO = 2), only χ -Fe₅C₂ (2.3–37.3%) were formed as active phases in the catalysts. Their TOFs in steady-state reactions were reported to be 0.038 s⁻¹. Cano et al.⁶³ evaluated an SBA-15-supported iron catalyst at 430 °C, 0.1 MPa, and H₂/CO = 2. The working catalyst contained 12% χ -Fe₅C₂ and 62% Fe₃O₄. On the basis of the results, the TOF of χ -Fe₅C₂ was calculated as 0.013 s⁻¹. Different from the oxidic supports, carbon supports are relatively inert to the active phases. Therefore, carbon-supported iron catalysts mainly consist of iron carbides and

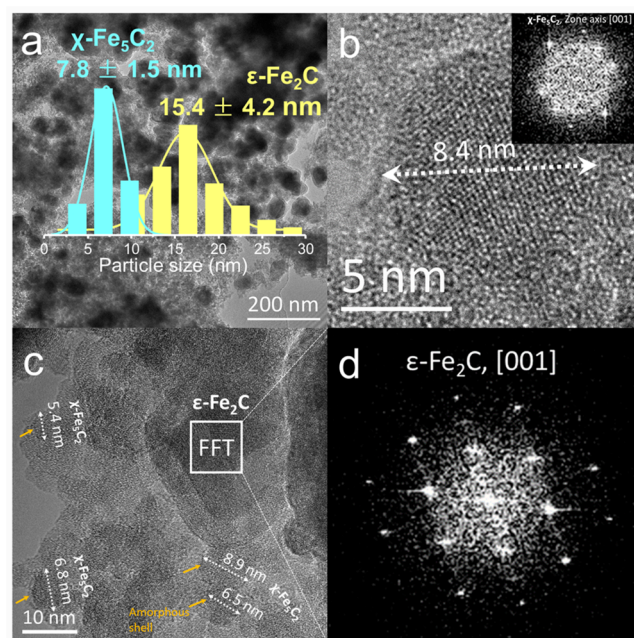


Figure 8. TEM of the spent FeSi-CO-H₂ catalyst. The PSD of the catalyst is inset in (a). The inset in (b) is the FFT image of the HRTEM image. (d) FFT image of the corresponding selected area with the plot marked FFT in (c).

exhibit the intrinsic FTS activity of iron carbides. Torres Galvis et al.^{61,65} reported a series of carbon nanofiber (CNF)-supported iron catalysts with a carburization degree of ~80%. The catalysts had apparent TOFs of 0.01–0.08 s⁻¹ under FTS conditions of 2.0 MPa, 340 °C, and H₂/CO = 1. Santos et al.⁶⁶ prepared a series of Fe@C catalysts using a MOF mediated synthesis method. The catalysts consisted of ~90% χ -Fe₅C₂ active phase and had apparent TOFs of 0.07–0.11 s⁻¹ under FTS conditions of 2.0 MPa, 340 °C, and H₂/CO = 1. Considering the differences in reaction conditions, the χ -Fe₅C₂ in the present work exhibits very consistent TOF values with those in the literature. Very little research has been reported about the intrinsic activity of ϵ -Fe₂C. Xu et al.¹² successfully prepared ϵ -Fe₂C with high carburization degree (ca. 73%) and found that it exhibited superior activity during low-temperature FTS (150 °C, 3 MPa, H₂/CO = 2). The TOF value was estimated to be 0.03 s⁻¹. Wezendonk et al.⁶⁷ found that the Fe MOF precursor mainly transformed to ϵ -Fe₂₂C when using a lower pyrolysis temperature of the MOF, and the TOF was reported to be ~0.06 s⁻¹ under FTS conditions of 340 °C, 2.0 MPa, and H₂/CO = 0.5. These values are of the same magnitude as our results. As for Fe₇C₃, there was no specified study to identify its FTS intrinsic activity in previous reports.²⁰ All in all, the crucial benchmark χ -Fe₅C₂ in the present study exhibits a consistent intrinsic activity with other reports. On the basis of that, the comparison of intrinsic activities of the three carbides (χ -Fe₅C₂, ϵ -Fe₂C, and Fe₇C₃) is credible, and Fe₇C₃ is found to be a distinctly active phase for FTS reaction.

The product selectivity is also distinct for each catalyst. Especially, the FeSi-H₂ catalyst exhibits the lowest CH₄ and highest C₅₊ selectivity. It has been reported that CO conversion could affect the selectivity because the increased concentration of water at higher CO conversion brought about the intensification of the WGS reaction at about the same rate as the FTS, leading to a high H₂/CO ratio and an increase in CH₄

selectivity and a decrease in longer hydrocarbon selectivity.⁶⁸ However, many researchers did not observe this correlation. Our previous works^{33,69} suggested that there is no direct correlation between the CO conversion and the hydrocarbon selectivity at moderate CO conversion levels in the Fe/SiO₂ catalyst. In the present work, we adjust the CO conversion of FeSi-H₂ and FeSi-syn catalysts to a comparative level with FeSi-CO catalyst (Table 1) and find that the selectivity is nearly not affected by CO conversion. Therefore, the influence of CO conversion on selectivity can be overlooked. It is also reported that the size can influence the iron phase structures and selectivity.^{61,70} For example, Raupp et al.⁴⁴ considered that larger particles tended to form χ -Fe₅C₂, whereas smaller ones formed a less stable ϵ' -carbide. McDonald et al.⁷¹ concluded that ϵ' -Fe_{2.2}C was easier to form in small particles than χ -Fe₅C₂, and small particles exhibited lower H₂ adsorption uptake. Wezendonk et al.⁶⁷ found that the carbide phase evolved in a sequence of ϵ -Fe_{2.2}C $\rightarrow$ χ -Fe₅C₂ $\rightarrow$ θ -Fe₃C with the particle size increasing from 2.5 to 28.4 nm when the calcination temperature increased from 400 to 900 °C during pyrolysis of Fe-BTC. In earlier work from our group,³³ the effect of size on selectivity was observed: the CH₄ selectivity gradually decreased and C₅₊ gradually increased with decreasing particle size. Pérez De Berti et al.³⁹ considered that more facets could be exposed when the particle size was small, and some of the emerging facets with a specific configuration could favor chain growth. The de Jong group⁶¹ studied the size effect in Fischer–Tropsch to lower olefins at 340–350 °C, H₂/CO = 1, and pressures of 1 and 20 bar. They found that methane and lower olefin selectivities were not affected on unpromoted catalysts when the size decreased from 7 to 2 nm. They presumed that methane formation takes place at highly active low coordination sites residing at corners and edges, which are more abundant on small iron carbide particles, and lower olefins are produced at promoted (stepped) terrace sites that are available and active, quite independent of size. The particle size of calcined catalyst in the present study is small (5.5 nm), and therefore, carbon-rich carbides (ϵ -Fe₂C and Fe₇C₃) are formed. However, the size-dependent selectivity is not observed in the present catalysts. Many studies indicated that the critical size for FTS selectivity is about 6 nm.^{62,70} The carbide particle sizes involved in the spent catalysts have been beyond this critical size. Therefore, the difference in selectivity mainly results from different features of carbides. The lowest CH₄ selectivity is accompanied by ϵ -Fe₂C, suggesting that ϵ -Fe₂C has outstanding C–C coupling ability. This is in agreement with previous works.^{53,71,72} McDonald et al.⁷¹ found that the fraction of C₂₊ increased with increasing content of ϵ' -carbides. Pour et al.⁷² suggested that the order of Fe–C bond strength is ϵ' -carbides < χ -carbides, and the surface carbons on the ϵ' -carbides are more easily hydrogenated to form hydrocarbons. Chun et al.⁵³ observed that the high ϵ'/χ carbide ratio corresponded to high C₅₊ productivity. In the present study, ϵ -Fe₂C exhibited superior selectivity compared to χ -Fe₅C₂ and Fe₇C₃.

4. CONCLUSIONS

The FTS activity and selectivity of ϵ -Fe₂C, χ -Fe₅C₂, and Fe₇C₃ carbide phases in the Fe/SiO₂ catalysts have been studied. By changing the gaseous atmospheres (syngas, CO, or H₂), at a moderate carburization degree pretreatment, three catalysts with different carbide phases are prepared from the same catalyst precursor. The identification of the carbide phases

shows that syngas, CO, and H₂ pretreatments lead to χ -Fe₅C₂, χ -Fe₅C₂ + Fe₇C₃, and χ -Fe₅C₂ + ϵ -Fe₂C as the carbide phases in the catalyst. The overall FTS activities for these catalysts are in the sequence of χ -Fe₅C₂ + Fe₇C₃ > χ -Fe₅C₂ + ϵ -Fe₂C > χ -Fe₅C₂. The activity is dependent on the carbide phase structure, phase content, and particle size. The structural properties of ϵ -Fe₂C, χ -Fe₅C₂, and Fe₇C₃ in each catalyst are systematically investigated by XRD, XAFS, MES, and TEM. The intrinsic activity of χ -Fe₅C₂ is calculated on the catalyst containing solely χ -Fe₅C₂ and is used as a benchmark for the calculation of ϵ -Fe₂C and Fe₇C₃. The precise discrimination of the intrinsic activities of χ -Fe₅C₂, Fe₇C₃, and ϵ -Fe₂C is achieved. It is found that Fe₇C₃ possesses the highest intrinsic activity (TOF = 4.59 $\times 10^{-2}$ s⁻¹) among the three carbides in the present Fe/SiO₂ catalyst. The different iron carbide also exhibits distinct influence on the product selectivity. The present work suggests that ϵ -Fe₂C produced lower CH₄ and higher C₅₊ selectivity than χ -Fe₅C₂ and Fe₇C₃. These findings under the MTFT reaction conditions may shed light on the mechanism of iron-catalyzed FTS reactions and provide new ideas for the development of FTS catalysts with a highly active or selective carbide phase as the predominant component.

■ ASSOCIATED CONTENT

§ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscatal.7b04085.

Elemental analysis results of the fresh calcined catalyst; XRD spectra, MES spectra, MES fitting results, EXAFS spectra, and EXAFS fitting results of reduced catalysts; MES spectra, STEM elementary mapping, and HRTEM images of spent catalysts; XRD and MES spectra of carbide standards (PDF)

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Notes

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Supporting information

Relationship between iron carbide phases (ϵ -Fe₂C, Fe₇C₃, and χ -Fe₅C₂) and catalytic performances of Fe/SiO₂ Fischer-Tropsch catalyst

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Table S1 XRF results of the calcined catalyst.

No.	Component	Content (mass %)	Measured elemental line
1	Fe ₂ O ₃	72.6	Fe-K α
2	SiO ₂	27.3	Si-K α (FA)
3	Na ₂ O	0.0010	Na-K α (FA)
4	MgO	0.0090	Mg-K α (FA)
5	Al ₂ O ₃	0.0132	Al-K α (FA)
6	P ₂ O ₅	0.0056	P-K α
7	SO ₃	0.0022	S-K α
8	Cl	0.0029	Cl-K α (FA)
9	K ₂ O	0.0057	K-K α
10	CaO	0.0167	Ca-K α
11	Cr ₂ O ₃	0.0196	Cr-K α
12	MnO	0.0180	Mn-K α
13	NiO	0.0195	Ni-K α
14	CuO	0.0148	Cu-K α
15	ZnO	0.0120	Zn-K α
16	ZrO ₂	0.0066	Zr-K α

Table S2 ICP-AES results of the calcined catalyst.

Element	Content
Fe	50.7%
Si	12.7%

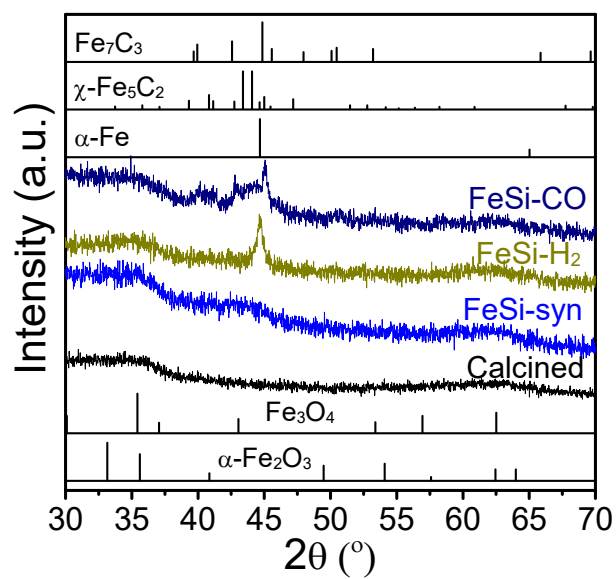


Figure S1 XRD patterns of the reduced catalysts after different atmospheres at 300 °C for 20 h.

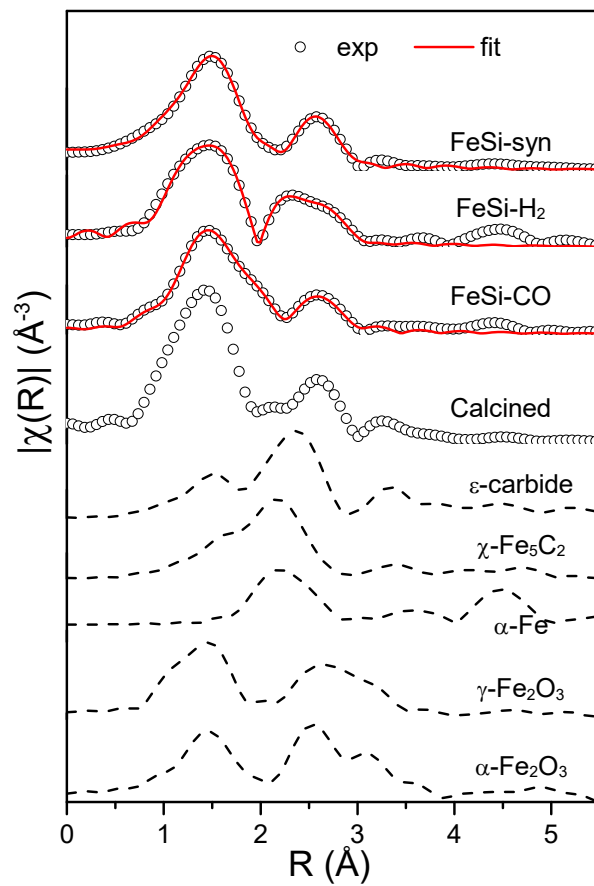


Figure S2 Fourier-transformed EXAFS spectra of the reduced catalysts after different atmospheres at 300 °C for 20 h.

Table S3 Results of EXAFS fitting of the reduced catalysts after different atmospheres at 300 °C for 20 h.

Catalysts	Shell	N	R (Å)	σ^2 (Å ²)	ΔE (eV)
FeSi-H ₂	Fe-O	4.9	1.72-2.29	0.020	2.9
	Fe-O	2.2	3.24	0.020	-3.2
	Fe-Fe	11.1	2.31-3.38	0.020	-3.2
FeSi-CO	Fe-O	1.5	1.93-2.10	0.0028	9.1
	Fe-Fe	0.9	3.03	0.0028	0.2
	Fe-C	1.3	1.80	0.0039	-9.1
	Fe-Fe	2.5	2.20-2.42	0.0039	-9.1
FeSi-syn	Fe-O	5	1.99	0.013	1.4
	Fe-Fe	7.5	3.12	0.020	7.0
	Fe-C	0.1	2.15	0.0080	11.0
	Fe-Fe	0.2	2.54-2.72	0.0080	11.0

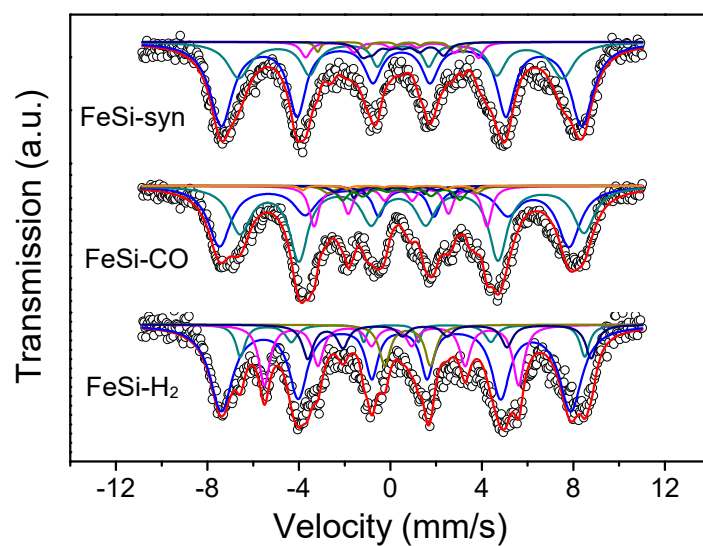


Figure S3 MES spectra of the reduced catalysts measured at 11 K.

Table S4 Iron phase compositions of the reduced catalysts from MES at 11 K.

Reduced catalysts	IS (mm/s)	QS (mm/s)	Hhf (kOe)	Phase	Compositions (%)
FeSi-H ₂	0.32		475	Fe ₃ O ₄ (A)	
	0.50		468	Fe ₃ O ₄ (B1)	74.6
	1.20		385	Fe ₃ O ₄ (B2)	
	0.75	2.0		Fe ²⁺	6.4
	0.05		345	α -Fe	19.0
FeSi-CO	0.40		475	Fe ₃ O ₄ (A)	
	0.65		464	Fe ₃ O ₄ (B)	64.9
	0.16		225	χ -Fe ₅ C ₂ (I)	
	0.00		187	χ -Fe ₅ C ₂ (II)	19.7
	0.16		130	χ -Fe ₅ C ₂ (III)	
	0.32		163	Fe ₇ C ₃ (I)	
	0.40		190	Fe ₇ C ₃ (II)	15.4
	0.15		208	Fe ₇ C ₃ (III)	
FeSi-syn	0.12		229	Fe ₇ C ₃ (IV)	
	0.49		490	Fe ₃ O ₄ (A)	83.9
	0.50		440	Fe ₃ O ₄ (B)	
	0.34		235	χ -Fe ₅ C ₂ (I)	
	0.40		198	χ -Fe ₅ C ₂ (II)	16.1
	0.40		110	χ -Fe ₅ C ₂ (III)	

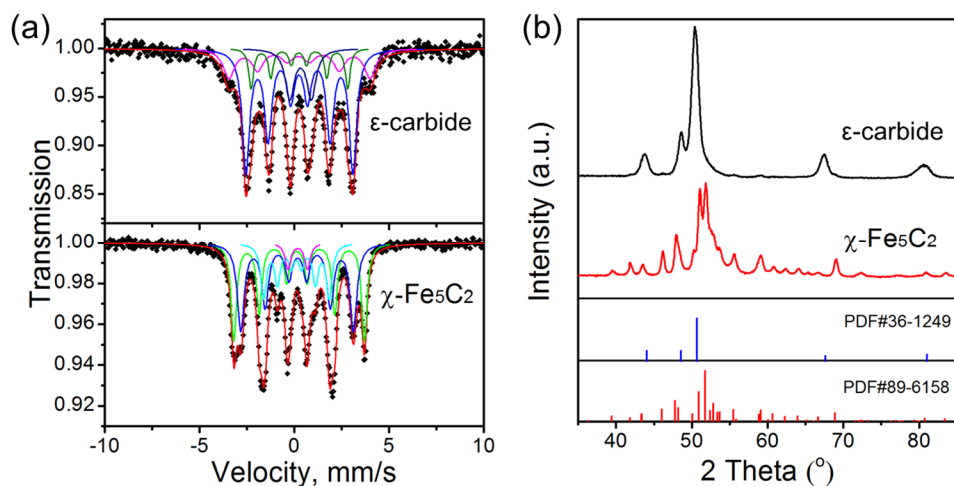


Figure S4 MES and XRD spectra of ϵ -carbide and χ -Fe₅C₂ carbide standards. ϵ -carbide and χ -Fe₅C₂ standard carbides were prepared using the gas carburization method. First, the precursor α -Fe₂O₃ was reduced to α -Fe by H₂ (99.999%, 80 ml/min) at 300 °C for 24 h. The sample was carburized by 1:4 CO/H₂ at 180 °C for 48 h to synthesize ϵ -carbide, and χ -Fe₅C₂ was acquired by annealing ϵ -carbide under Ar (99.999%, 50 ml/min) at 350 °C.

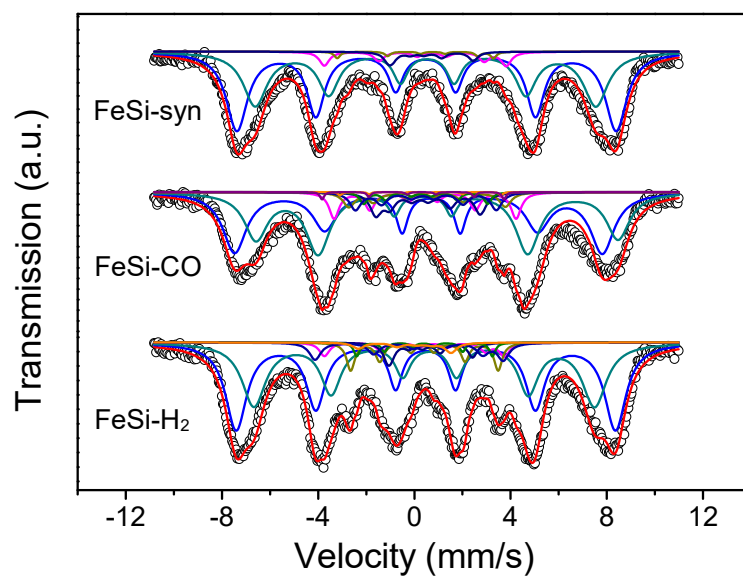


Figure S5 MES spectra of the spent catalysts measured at 11 K.

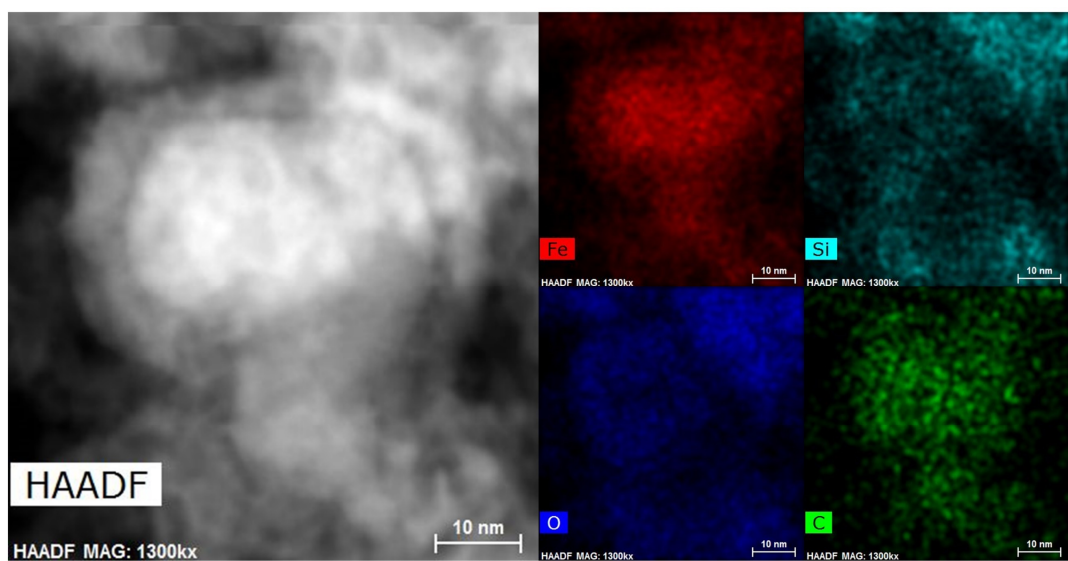


Figure S6 STEM elementary mapping of the spent FeSi-CO catalyst.

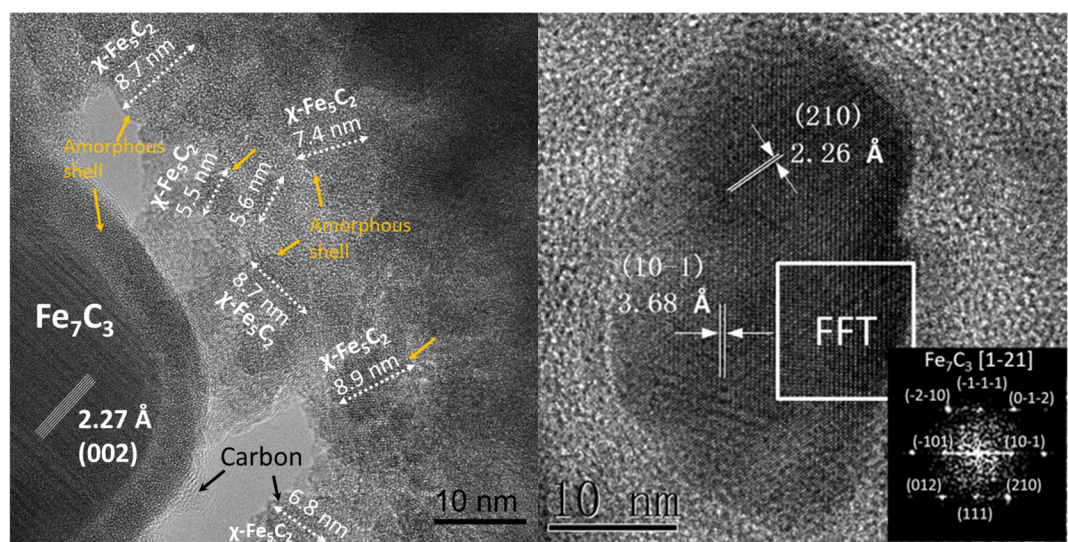


Figure S7 TEM of the spent FeSi-CO catalyst.

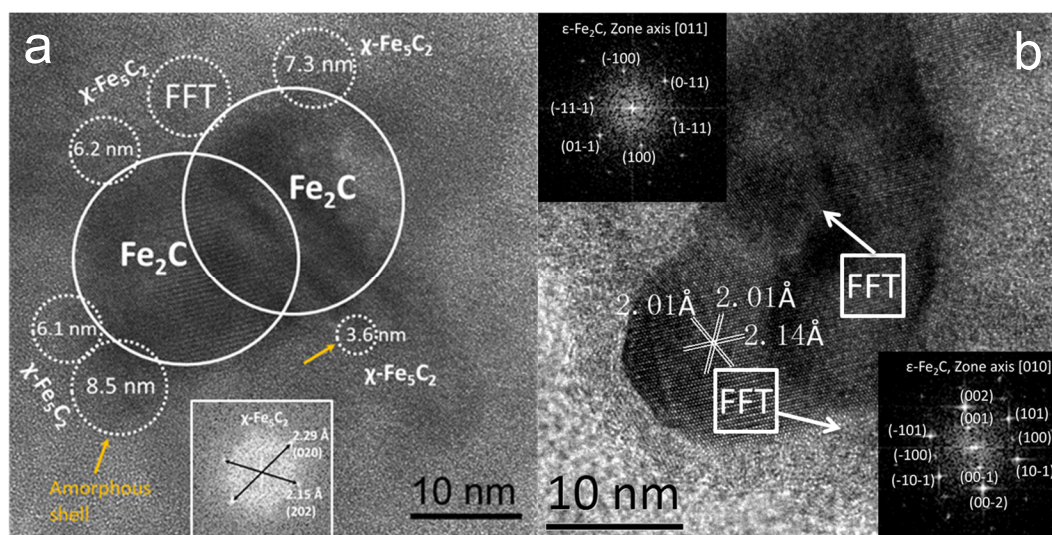


Figure S8 TEM of the spent FeSi-H₂ catalyst.

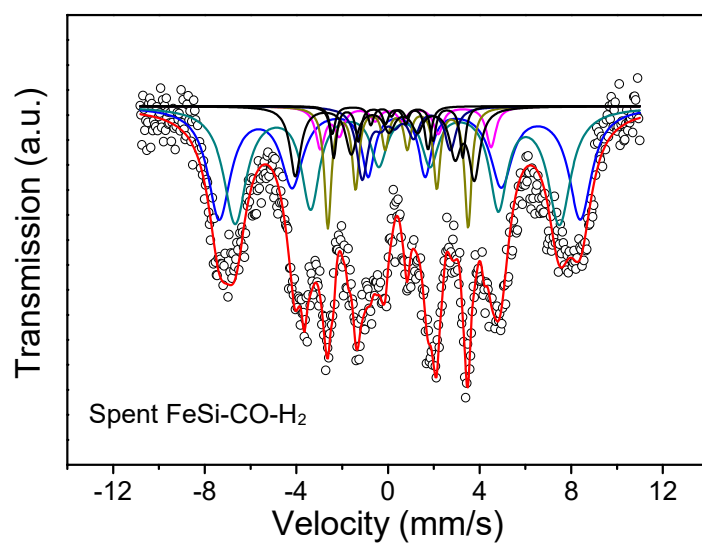


Figure S9 MES spectra of the spent FeSi-CO-H₂ catalyst measured at 11 K.

Table S5 Iron phase compositions of the spent FeSi-CO-H₂ catalyst from MES at 11 K.

Spent catalyst	IS (mm/s)	Hhf (kOe)	Phase	Compositions (%)
FeSi-CO-H ₂	0.45	490	Fe ₃ O ₄ (A)	59.6
	0.56	440	Fe ₃ O ₄ (B)	
	0.40	232	χ -Fe ₅ C ₂ (I)	
	0.40	190	χ -Fe ₅ C ₂ (II)	15.3
	0.40	120	χ -Fe ₅ C ₂ (III)	
	0.39	175	ϵ -Fe ₂ C (I)	
	0.27	243	ϵ -Fe ₂ C (II)	22.7
	0.07	131	ϵ -Fe ₂ C (III)	

Table S6 Comparison of TOFs between this work and literature.

Sample	Reaction conditions	Iron phase	Content in iron	TOF (s ⁻¹)	Reference
FeSi	260 °C, 3.0 MPa, H ₂ /CO=2	χ -Fe ₅ C ₂	11.0-16.5%	0.0162	This work
FeSi	260 °C, 3.0 MPa, H ₂ /CO=2	ϵ -Fe ₂ C	14.7-22.7%	0.0150	This work
FeSi	260 °C, 3.0 MPa, H ₂ /CO=2	Fe ₇ C ₃	16.3%	0.0459	This work
Fe/xSiO ₂	280 °C, 1.5 MPa, H ₂ /CO=2	χ -Fe ₅ C ₂	2.3-37.3%	0.038	33
Fe/SBA-15	430 °C, 0.1 MPa, H ₂ /CO=2	χ -Fe ₅ C ₂	12%	0.013	63
Fe/CNF	350 °C, 0.1 MPa, H ₂ /CO=2	χ -Fe ₅ C ₂	50-80%	0.002-0.012	61
Fe/CNF	340 °C, 2.0 MPa, H ₂ /CO=1	χ -Fe ₅ C ₂	~80%	0.01-0.08	61
Fe@C	340 °C, 2.0 MPa, H ₂ /CO=1	χ -Fe ₅ C ₂	~90%	0.07-0.11	66
RQ Fe	150 °C, 3 MPa, H ₂ /CO=2	ϵ -Fe ₂ C	73%	0.03	12
Fe@C-450	340 °C, 2.0 MPa, H ₂ /CO=0.5	ϵ -Fe _{2.2} C	53%	0.06	67